SUBMITTAL TRANSMITTAL FORM

MAGUIRE IRON, INC
PO BOX 1446
1610 N MINNESOTA AVE
SIOUX FALLS, SD 57104
PHONE: (605) 334-9749
FAX: (605) 334-9752

ENGINEER: Banner Associates, Inc.
OWNER: Minnehaha Community Water Corp.
PROJECT: 250,000 Gallon SPHERE
DATE: 2/22/2022

SUBMITTAL NO. : HUM001

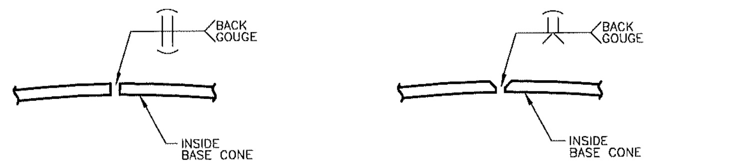
SUBMITTAL DESCRIPTION
STRUCTURAL DESIGN - with Calcs

CONTRACTOR'S CERTIFICATION:

Contractor has determined or verified all quantities dimensions field construction criteria, materials catalog numbers and similar data, and has coordinated the information within the submittal with the require-ments of the contract documents, and assumes full responsibility for so doing.

MAGUIRE IRON, INC

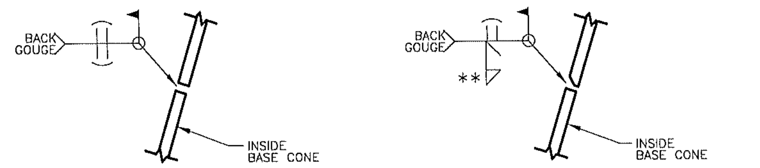
By: **Kaleb Johnson** **Date** 2/22/2022



THIS IS A TYPICAL DETAIL OF ALL VERTICAL WELDS IN BASE CONES WHERE THICKNESS IS LESS THAN OR EQUAL TO 3/8".

THIS IS A TYPICAL DETAIL OF ALL VERTICAL WELDS IN BASE CONES WHERE THICKNESS IS GREATER THAN 3/8".

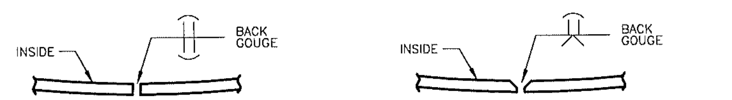
1 E1 DETAIL



THIS IS A TYPICAL DETAIL OF ALL HORIZONTAL WELDS IN BASE CONES WHERE THICKNESS IS LESS THAN OR EQUAL TO 3/8".

THIS IS A TYPICAL DETAIL OF ALL HORIZONTAL WELDS IN TANK AND BASE CONE WHERE THICKNESS IS GREATER THAN 3/8".

2 E1 DETAIL



THIS IS A TYPICAL DETAIL OF ALL VERTICAL WELDS IN TANK AND PEDESTAL WHERE THICKNESS IS LESS THAN OR EQUAL TO 3/8"

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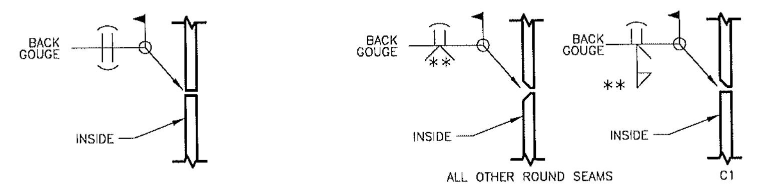
3 E1 DETAIL



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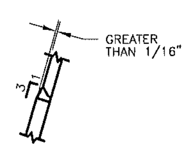
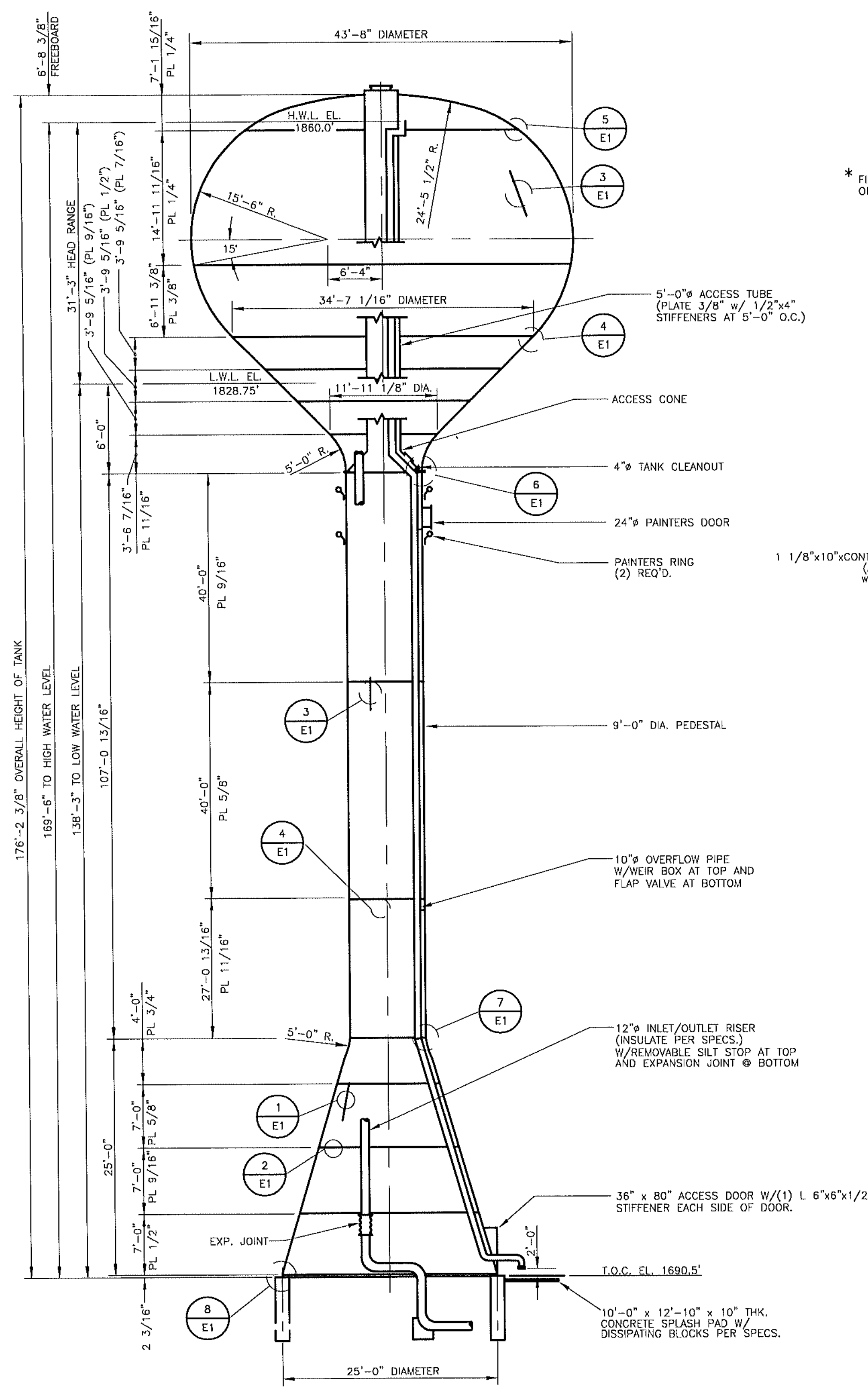
** WHEN THE DIFFERENCE IN THICKNESSES BETWEEN ADJACENT HORIZONTALLY WELDED PLATES EXCEEDS 1/16", A 3:1 TAPERING SHALL BE PERFORMED BETWEEN THE PLATES. THIS TAPERING MAY BE ACHIEVED BY TAPERING THE WELD OR BY TAPERING THE THICKER PLATE.



4 E1 DETAIL

REGISTERED PROFESSIONAL ENGINEER REG. NO. 6353 JOHN P. LONGNECKER SOUTH DAKOTA

2/2/22



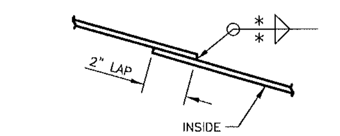
ELEVATION

TANK DESIGN DATA

CAPACITY	250,000 GALLONS
SNOW LOAD	40 PSF
WIND LOAD	90 MPH
SEISMIC	AWWA D100-11, IBC
STANDARD	AWWA D100-11, IBC
MATERIALS	STEEL PLATE-ASTM A36 ALL MATERIALS PER SECTION 2 OF AWWA-D-100-11
CORROSION ALLOWANCE	NONE
LIGHTING, FIXTURES, CONTROLS	PER PROJECT SPECIFICATIONS
RADIOGRAPHIC EXAMINATION	SPOT X-RAY PER AWWA D100-11
PAINTING AND DISINFECTION	PER PROJECT SPECIFICATIONS
INSULATION	PER PROJECT SPECIFICATIONS
TESTING	PER AWWA D100-11
PROJECT SPECIFICATIONS	SPECIFICATION FOR 250,000 GALLON ELEVATED WATER STORAGE TANK FOR THE CITY OF HUMBOLT, SD BY BANNER WITH GEOTECHNICAL REPORT BY GEOTEK PSI DATED: MARCH, 2021

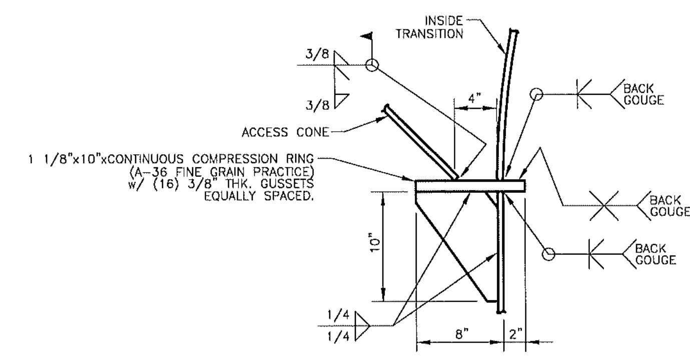
ALL BUTT WELDED JOINTS SUBJECT TO PRIMARY STRESSES THAT ARE DUE TO WATER PRESSURE OR WEIGHT MUST BE FULL PENETRATION AND FULL FUSION.

ALL COMPRESSION AND TENSION RING RADIAL WELDS ARE TO BE FULL PENETRATION AND FULL FUSION.

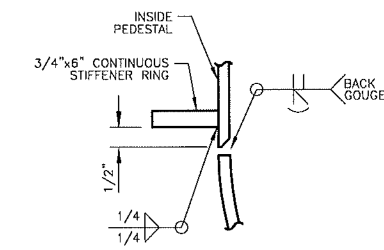


* FILLET WELD SIZE SHALL BE EQUAL TO THICKNESS OF THINNER OF MEMBERS JOINED.

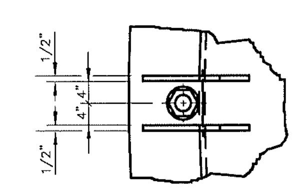
5 E1 DETAIL



6 E1 DETAIL



7 E1 DETAIL



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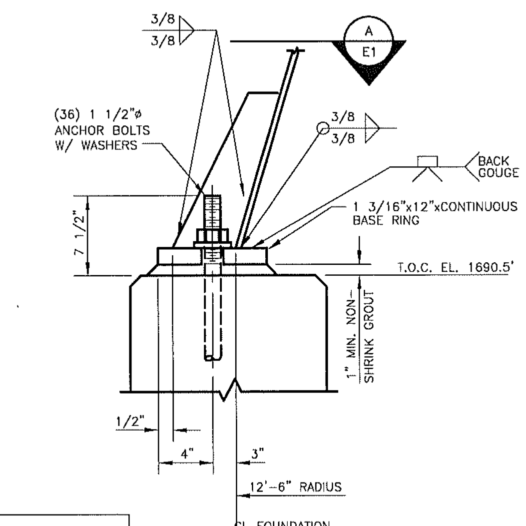


8 E1 DETAIL

CONTRACTOR'S CERTIFICATION:

Contractor has determined or verified all quantities, dimensions, field construction criteria, materials, catalog numbers, and similar data, and has coordinated the information within the submittal with the requirements of the contract documents, and assumes full responsibility for so doing.

MAGUIRE IRON, INC.

Date 02/22/2022

STRUCTURAL DESIGN

DATE 02-09-22

I HEREBY CERTIFY THAT THIS PLAN, SPECIFICATION OR REPORT WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY REGISTERED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF SOUTH DAKOTA

DATE: 2/4/22 REG.NO. 6353

NO.	DATE	DR. BY	CK. BY	DESCRIPTION

SCALE NONE	
DR. ZDL	DATE 02-07-22
CK. JPL	DATE 02-07-22

River Valley Engineering, Inc.
(479) 410-2208
Fax : (479) 410-2440
111 North Plaza Ct.
Van Buren, AR 72956

Maguire Iron, Inc.
GENERAL CONTRACTOR
SIOUX FALLS, SOUTH DAKOTA

TANK ELEVATION AND DETAILS
250,000 GALLON SPHEROID
MCWC-HUMBOLT, SD

RVE DWG. FILE NAME	:\MAGUIRE\2022-789\2022-789E1.DWG	
SHEET 1	OF 3	RVE PROJECT NO. 2022-789
DRAWING NO.	2022-789E1	

NORTH

ANCHOR BOLT CHORD DIMENSIONS			
1 TO 2	1 TO 3	1 TO 4	1 TO 5
2'-2 11/16"	4'-5 1/8"	6'-7 3/16"	8'-8 11/16"

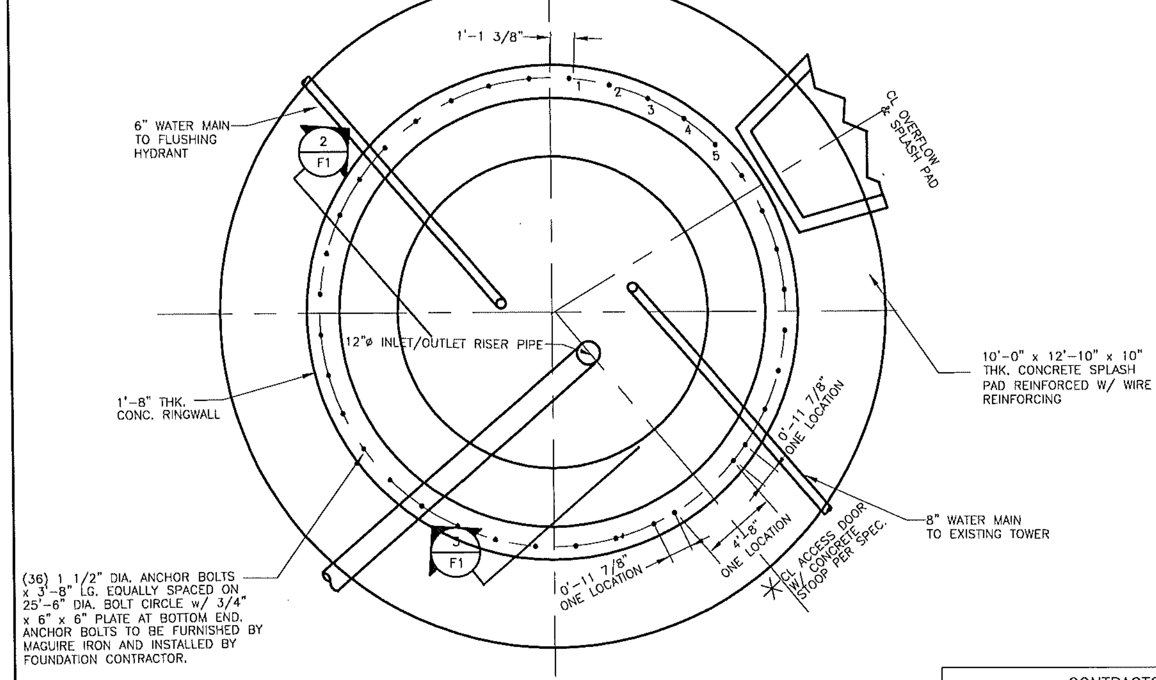
PANEL BOX, ELECTRIC BRACKETS AND ELECTRICAL PIPE SLEEVES ARE TO BE LOCATED AND IDENTIFIED BY PROJECT ENGINEER, CONTRACTOR AND APPLICABLE SUBCONTRACTORS.

FOUNDATION CONTRACTOR TO PROVIDE SLEEVES IN FOUNDATION RINGWALL FOR POWER SUPPLY LINE AND TELEPHONE LINE. COORDINATE WITH ELECTRICAL CONTRACTOR AND TELEPHONE COMPANY FOR SIZE AND LOCATION.

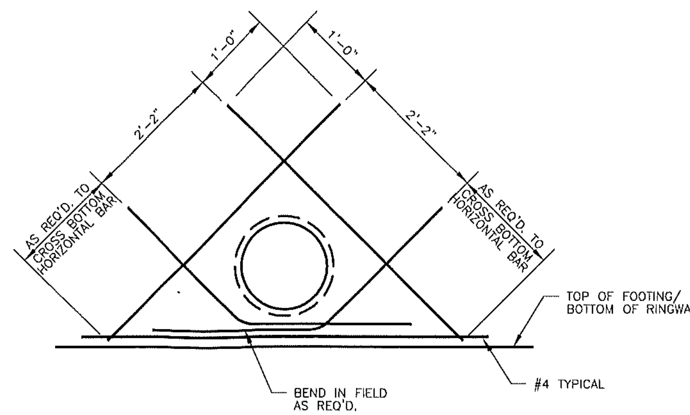
FOUNDATION NOTES

1. ALL FOUNDATION WORK SHALL BE PERFORMED IN ACCORDANCE WITH ACI 318-2014 AND SPECIFICATION FOR 250,000 GALLON ELEVATED WATER STORAGE TANK FOR THE CITY OF HUMBOLT, SD BY BANNER WITH GEOTECHNICAL REPORT BY GEOTEK PSI DATED: MARCH, 2021
2. ALL CONCRETE WORK SHALL BE PERFORMED IN ACCORDANCE WITH ACI 318-2014.
3. DESIGN AND TOLERANCES SHALL BE PER AWWA D100-11 AND ACI 318-2014.
4. ALLOWABLE NET SOIL BEARING PRESSURE FOR WATER, SNOW AND DEAD LOADS=5,000 PSF.
5. ALLOWABLE NET SOIL BEARING PRESSURE FOR WATER, DEAD AND WIND LOADS=6,667 PSF.
5. ALLOWABLE NET SOIL BEARING PRESSURE FOR WATER, DEAD AND SEISMIC LOADS=6,667 PSF.
6. REINFORCING STEEL: ASTM A615 GR. 60. (Fy=60,000)
7. REBAR SPLICE SHALL COMPLY WITH ACI 318R-2014 w/MIN. OF 42 BAR DIA.
8. CONCRETE SHALL HAVE 28 DAY COMPRESSIVE STRENGTH OF 4000 PSI.
9. ALL EXPOSED CONCRETE EDGES SHALL HAVE A 1"x45° CHAMFER.
10. ALL EXCAVATION, BACKFILL AND LEVELING SHALL BE PERFORMED PER ABOVE REFERENCED SPEC.

BANNER TO VERIFY ITEMS MARKED WITH (*)



1 F1 FOUNDATION PLAN



4 F1 DETAIL OF EXTRA REINFORCING AROUND WALL SLEEVE

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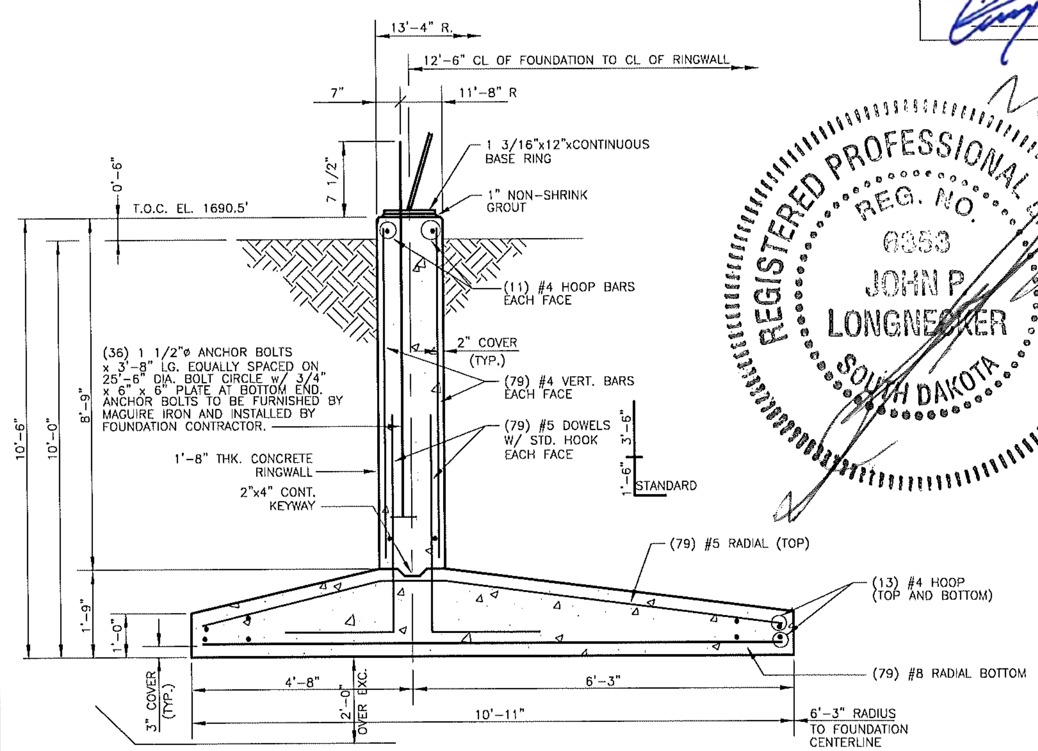
MAGUIRE IRON, INC.

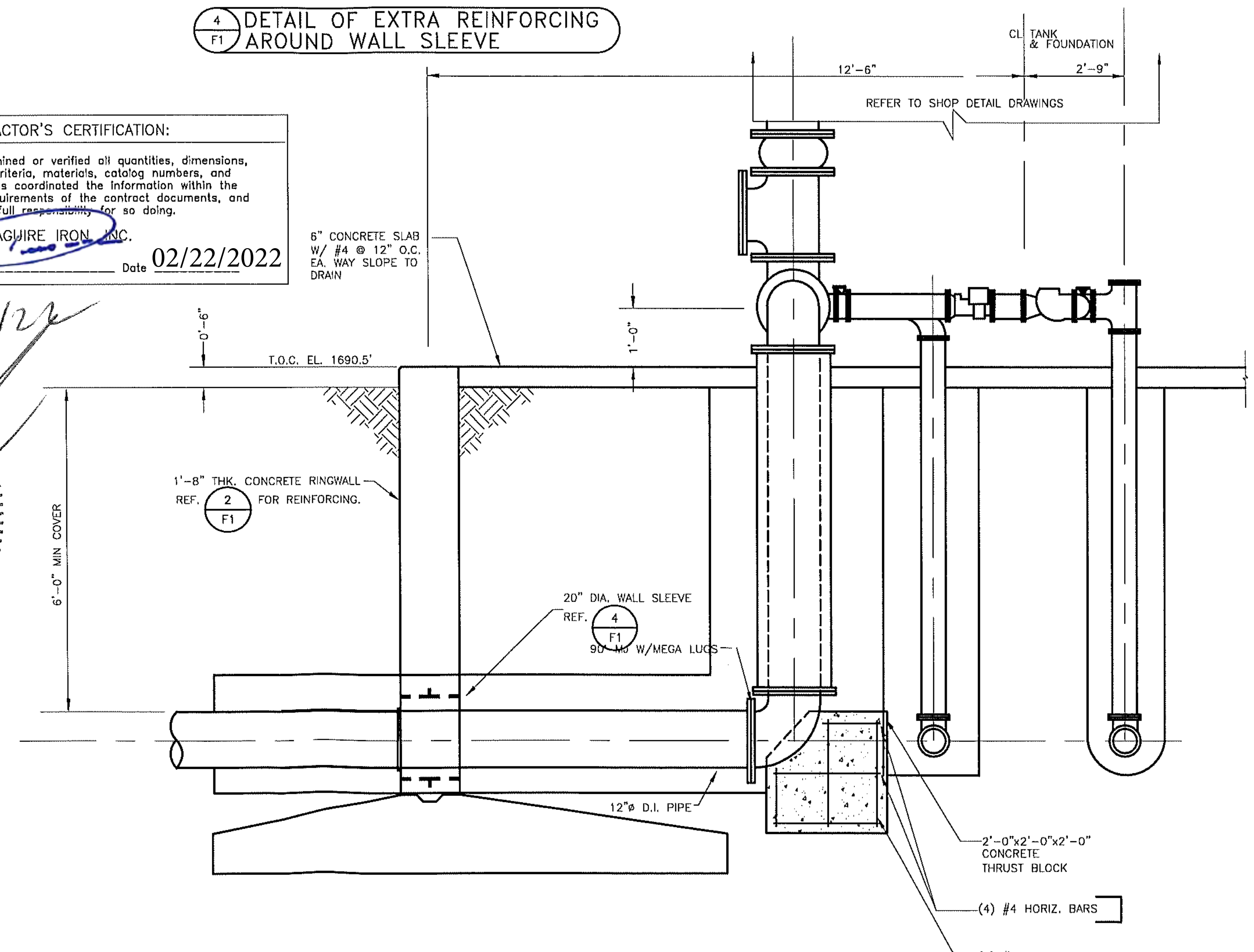
Date 02/22/2022

REGISTERED PROFESSIONAL ENGINEER REG. NO. 6353 JOHN P. LONGNECKER SOUTH DAKOTA

2/21/22



2 F1 SECTIONAL ELEVATION THROUGH RINGWALL



3 F1 SECTIONAL ELEVATION AT RISER PIPE

STRUCTURAL DESIGN

DATE 02-09-22

ESTIMATED FOUNDATION QUANTITIES		
ITEM	CONCRETE (CU. YDS.)	STEEL (LBS.)
RINGWALL	43	3200
FOOTING	44	4700
FLOOR	8	700
PILE CAP		
PILE		
TOTAL	95	8,600

I HEREBY CERTIFY THAT THIS PLAN, SPECIFICATION OR REPORT WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY REGISTERED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF SOUTH DAKOTA

DATE: 2/21/22 REG.NO.: 6353

NO.	DATE	DR. BY	CK. BY	DESCRIPTION

SCALE NONE	
DR. ZDL	DATE 02-07-22
CK. JPL	DATE 02-07-22

River Valley Engineering, Inc.
(479) 410-2208
Fax: (479) 410-2440
111 North Plaza Ct.
Van Buren, AR 72956

Maguire Iron, Inc.
GENERAL CONTRACTOR
SIOUX FALLS, SOUTH DAKOTA

FOUNDATION PLAN AND DETAILS
250,000 GALLON SPHEROID
MCWC-HUMBOLT, SD

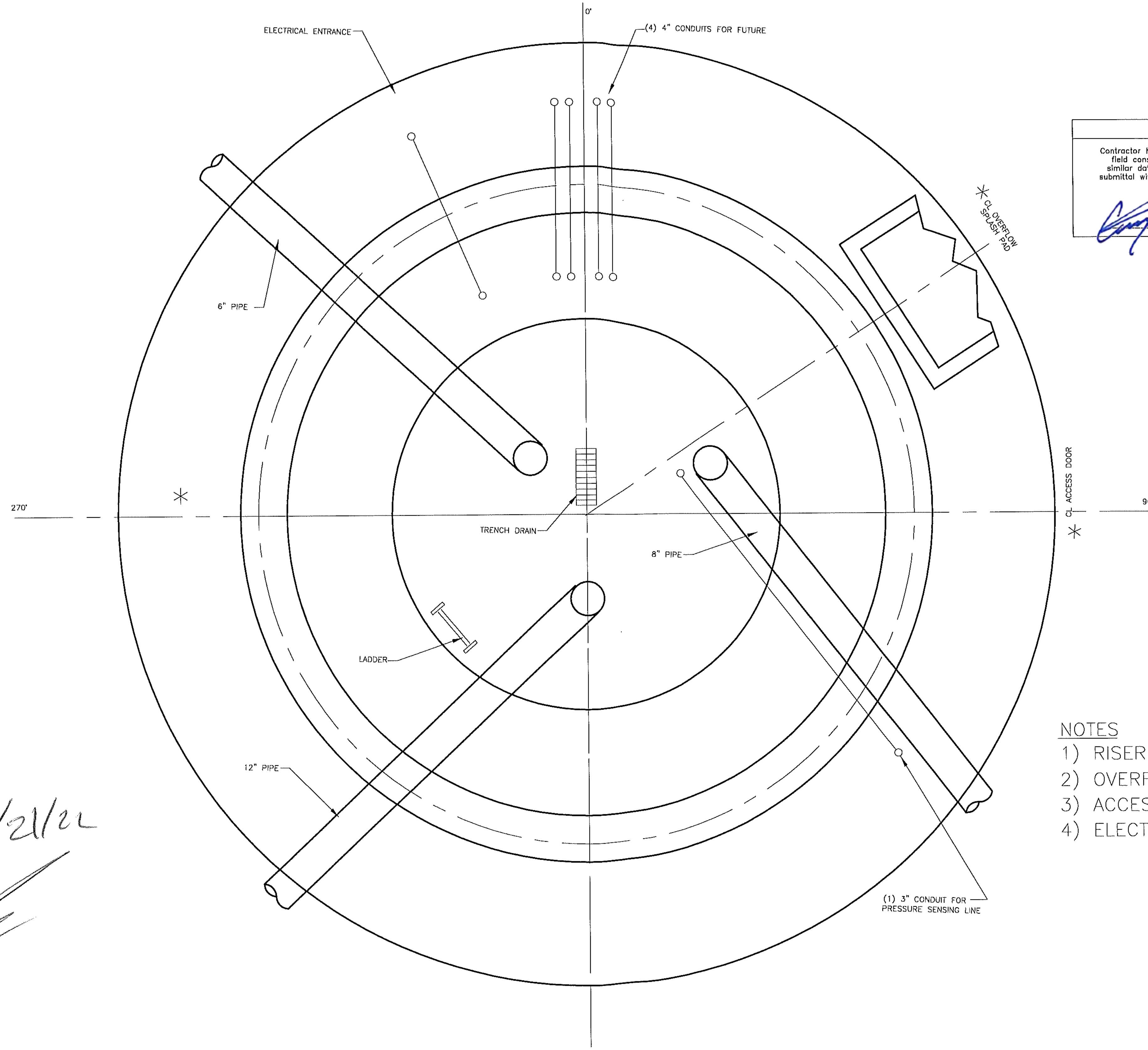
RVE DWG. FILE NAME: C:\MAGUIRE\2022-789\2022-789F1.DWG

SHEET 2 OF 3

RVE PROJECT NO. 2022-789

DRAWING NO. 2022-789F1

NORTH



BASE ORIENTATION

CONTRACTOR'S CERTIFICATION:

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MAGUIRE IRON, INC.

Date 02/22/2022

NOTES

1) RISER AT (225°)
2) OVERFLOW AND SPLASHPAD AT (58°)
3) ACCESS DOOR AT (135°)
4) ELECTRIC ENTRANCE AT (315°)

REGISTERED PROFESSIONAL ENGINEER
REG. NO.
6353
JOHN P.
LONGNECKER
SOUTH DAKOTA

2/21/22

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DATE: 2/21/22 REG.NO.: 6353

07FEB22

NO.	DATE	DR. BY	CK. BY	DESCRIPTION

SCALE NONE	
DR. ZDL	DATE 02-07-22
CK. JPL	DATE 02-07-22

River Valley Engineering, Inc.
(479) 410-2208
Fax: (479) 410-2440
111 North Plaza Ct.
Van Buren, AR 72956

Maguire Iron, Inc.
GENERAL CONTRACTOR
SIOUX FALLS, SOUTH DAKOTA

TANK ORIENTATION
250,000 GALLON SPHEROID
MCWC-HUMBOLT, SD

STRUCTURAL DESIGN

DATE 02-09-22

RVE DWG. FILE NAME C:\MAGUIRE\2022-789\2022-789E1.DWG	
SHEET 3 OF 3	RVE PROJECT NO. 2022-789
DRAWING NO. 2022-78901	

RVE

River Valley Engineering, Inc.

111 North Plaza Court, Van Buren, AR 72956
(479) 410-2208 Fax (479) 410-2440

Project No. :	2022-789	Subject :	MCWC-Humboldt, SD	Page :
Calc. by :	JPL	Date :	2/10/2022	Rev :

Calculations

FOR

250,000 GALLON SPHEROID

WATER STORAGE TANK

MCWC-HUMBOLDT, SOUTH DAKOTA

2/10/22

REGISTERED PROFESSIONAL ENGINEER
REG. NO.
6953
JOHN P.
LONGNECKER
SOUTH DAKOTA

SPHEROID TANK DESIGN

(Tank Design Conforms to AWWA D-100-2011 Standard)

Project Number	2022-789
City and State	Dell Rapids, SD
Date / Time	1/17/2022 6:15:48 PM
Design Capacity (Gal)	250000
Working Capacity (Gal)	250132

DESIGN PARAMETERS

Height to High Water Line (ft)	169.5000
Head Range (ft)	31.2500
Free Board (ft)	6.6976
Snow Load (psf)	40
Wind Speed (mph)	90
Seismic Use Group	III
Site Class (A,B,C,D,E,F)	D
Seismic Importance Factor	1.5
Earthquake Response Acc. 0.2 Sec, Ss	0.1
Earthquake Response Acc. 1.0 Sec, S1	0.033
Longer-Period Ground Motion, TL	12
Short Period Site Coefficient, Fa	1.6
Long Period Site Coefficient, Fv	2.4
Damped Short Period Site Coef., SDS	0.106672
Damped Long Period Site Coef., SD1	0.05280264
Natural Period of Tower (Seconds)	4.0000
Equivalent Fixed Percentage (%)	0.5940
Iterations to Convergence	4
P-delta Moment due to Seismic (ft-kips)	1295.0385
P-delta Moment due to Wind (ft-kips)	1138.7034
Deflection Due to Seismic (in)	6.6464
Deflection Due to Wind (in)	5.5566

Corrosion Allowance

Wetted (in)	0.0000
Non-wetted (in)	0.0000

Design Notes:

1. Seismic and Wind Design is per Worst Case of AWWA D-100, IBC and ASCE-7 methods.
2. Specified Additional Shear Due to Antenna (kips) 0.0000
3. Specified Additional Moment Due to Antenna (ft-kips) 0.0000
4. Miscellaneous Load for platforms, piping etc. (kips) 15.0000

TANK GEOMETRY

Diameter (ft)	43.6660
Overall Height of Tank (ft)	176.1976
Radius of Roof (ft)	24.4562
Height of Roof (ft)	7.1631
Radius of Knuckle (ft)	15.5000
Offset of Knuckle (ft)	6.3330
Height of Top Knuckle (ft)	10.9602
Height of Bottom Knuckle (ft)	10.9602
Height of Upper Cone (ft)	11.3287
Diameter at Top of Upper Cone (ft)	34.5863
Radius of Transition (ft)	5.0000
Height of Transition (ft)	3.5355
Diameter at Top of Transition (ft)	11.9289
Diameter of Pedestal (ft)	9.0000
Height of Pedestal (ft)	107.0417
Diameter of Base Cone (ft)	25.0000
Height of Base Cone (ft)	25.0000
Top of Pedestal to Low Water Line (ft)	6.0000
Diameter of Access Tube (ft)	5.0000

Design of Tank Sections

Roof

Thickness (in)	0.2500
Weight of Roof (kips)	11.2271
Surface Area of Roof (sqft)	1100.6963
Height to Centroid of Plate (ft)	2.6861
Height to Centroid of Projected Area (ft)	2.6861
Volume of Water in Roof (cuft)	425.4314
Weight of Water in Roof (kips)	26.5469
Height to Centroid of Water (ft)	0.2327
Shear Due to Earthquake by Water (kips)	0.1577
Shear Due to Earthquake by Plate (kips)	0.0667
Moment Due to Earthquake by Water (ft-kips)	0.0367
Moment Due to Earthquake by Plate (ft-kips)	0.2389

Top Knuckle

Thickness (in)	0.2500
Weight of Top Knuckle (kips)	15.8285
Surface Area of Top Knuckle (sqft)	1551.8105
Height to Centroid of Plate (ft)	5.3864
Height to Centroid of Projected Area (ft)	5.2846
Volume of Top Knuckle (cuft)	14418.7491
Weight of Water in Top Knuckle (kips)	899.7299
Height to Centroid of Water (ft)	5.0974
Shear Due to Earthquake by Water (kips)	5.3444
Shear Due to Earthquake by Plate (kips)	0.0940
Moment Due to Earthquake by Water (ft-kips)	29.0073
Moment Due to Earthquake by Plate (ft-kips)	1.4762

Bottom Knuckle

Thickness (in)	0.3750
Weight of Bottom Knuckle (kips)	23.7427
Surface Area of Bottom Knuckle (sqft)	1551.8105
Height to Centroid of Plate (ft)	5.6755
Height to Centroid of Projected Area (ft)	2.6861
Volume of Bottom Knuckle (cuft)	14418.7491
Weight of Water in Bottom Knuckle (kips)	899.7299
Height to Centroid of Water (ft)	5.8628
Shear Due to Earthquake by Water (kips)	5.3444
Shear Due to Earthquake by Plate (kips)	0.1410
Moment Due to Earthquake by Water (ft-kips)	120.6441
Moment Due to Earthquake by Plate (ft-kips)	5.5695

DESIGN OF TANK SECTIONS (CONT'D)

Upper Cone

Design - Section	Thickness (in)	Weight (kips)	Height (ft)
1	0.4375	9.2269	3.7762
2	0.5000	7.9601	3.7762
3	0.5625	6.0471	3.7762

Transition

Thickness (in)	0.6875
Weight of Transition (kips)	3.4594
Surface Area of Transition (sqft)	123.3311
Height to Centroid of Plate (ft)	1.9518
Height to Centroid of Projected Area (ft)	1.8516
Volume of Transition (cuft)	274.7538
Weight of Water in Transition (kips)	17.1446
Height to Centroid of Water (ft)	1.9379
Shear Due to Earthquake by Water (kips)	0.1018
Shear Due to Earthquake by Plate (kips)	0.0149
Moment Due to Earthquake by Water (ft-kips)	303.0569
Moment Due to Earthquake by Plate (ft-kips)	5.8698

Access Cone

Thickness (in)	.5000
Weight of Access Cone (kips)	1.4242

Access Tube

Thickness of Bottom 15 Feet (in)	.3750
Thickness of Remaining Height (in)	.3750
Weight of Access Tube (kips)	9.7609

DESIGN OF TANK SECTIONS (CONT'D)

Pedestal

Design - Section	Thickness (in)	Weight (kips)	Height (ft)
1	0.5625	5.1912	8.0000
2	0.5625	10.3823	8.0000
3	0.5625	15.5735	8.0000
4	0.5625	20.7647	8.0000
5	0.5625	25.9558	8.0000
6	0.6250	31.7238	8.0000
7	0.6250	37.4918	8.0000
8	0.6250	43.2597	8.0000
9	0.6250	49.0277	8.0000
10	0.6250	54.7957	8.0000
11	0.6875	61.1404	8.0000
12	0.6875	67.4852	8.0000
13	0.6875	73.8299	8.0000
14	0.6875	76.2423	3.0417

Base Cone

Design - Section	Thickness (in)	Weight (kips)	Height (ft)
1	0.7500	4.1504	4.0000
2	0.6250	8.1253	7.0000
3	0.5625	9.6867	7.0000
4	0.5000	10.7206	7.0000

Base Plate

Base Plate Thickness (in) 1.1875

Note: Use a standard 1 in. of non-shrink grout under the base plate.

Anchor Bolt -- Mild Steel

Anchor Bolt Size (in)	Quantity Req'd	Tension Load (kip)	Minimum Embedment (in)
1.50	34	26334	58.91
1.75	25	35511	68.09
2.00	19	46683	78.32

Anchor Bolt With Plate Attached (Breakout Procedure)

Anchor Bolt Size (in)	Quantity Req'd	Tension Load(Ult) (kip)	Minimum Embedment (in)
1.50	34	48500	25.00
1.75	26	48500	25.00
2.00	26	48500	25.00

Anchor Bolt Chair

Anchor Bolt Size (in)	Top Plate Thickness (in)	Top Plate Width (in)	Side Plate Thickness (in)	Anchor Chair Height (in)
1.50	1.1250	9.0000	0.5000	12.0000
1.75	1.2041	9.0000	0.5000	12.0000
2.00	1.4068	9.0000	0.5000	12.0000

LOADS ON THE FOUNDATION

Loads From Plate Steel

Section of Tank	Weight (Kips)
Roof	11.2271
Top Knuckle	15.8285
Bottom Knuckle	23.7427
Upper Cone	23.2341
Transition	3.4594
Pedestal	76.2423
Base Cone	32.6830
Base Plate	4.0088
Access Shaft	9.7609
Access Cone	1.4242
Total Load From Plate Steel	201.6110
Loads From Miscellaneous	15.0000
Loads From Snow	37.5801
Loads From Water	2121.5838
Total Load On Foundation From Tank	2375.7749

Moments and Shears on the Foundation

Wind Moment at Top of Pedestal (ft-kips)	629.9610
Wind Shear at Top of Pedestal (kips)	27.4583
Seismic Moment at Top of Pedestal (ft-kips)	308.9267
Seismic Shear at Top of Pedestal (kips)	13.3280
Wind Moment at Top of Foundation (ft-kips)	6846.7275
Wind Shear at Top of Foundation (kips)	51.3387
Seismic Moment at Top of Foundation (ft-kips)	3403.7135
Seismic Shear at Top of Foundation (kips)	13.9750

RINGWALL FOUNDATION DESIGN

(Foundation Design Conforms to AWWA D-100-11 Specification)

Project Number	2022-789
City and State	Dell Rapids, SD
Date / Time	1/17/2022 6:19:05 PM
Tank Capacity (Gal)	250000

DESIGN PARAMETERS

Diameter of Ringwall (ft)	25.0000
Depth to Bottom of Footing (ft)	10.0000
Weight of Water (kip)	2122.0000
Weight of Steel (kip)	202.0000
Weight of Snow (kip)	38.0000
Wind Moment at Top of Ringwall (ft-kip)	6847.0000
Wind Shear at Top of Ringwall (kip)	52.0000
Seismic Moment at Top of Ringwall (ft-kip)	3404.0000
Seismic Shear at Top of Ringwall (kip)	14.0000
Compressive Strength of Concrete (ksi)	4.0000
Yield Strength of Reinforcement Steel (ksi)	60.0000
Unit Weight of Soil (pcf)	100.0000
Depth of Water Table (ft)	20.0000
Bearing Capacity for Dead Loads (ksf)	5.0000
Bearing Capacity for Wind Loads (ksf)	6.6700
Bearing Capacity for Seismic Loads (ksf)	6.6700
Coefficient of Passive Pressure	2.0000
Coefficient of Friction Between Fnd & Soil	0.4000

RINGWALL DESIGN

Diameter C-C of Ringwall (ft)	25.0000
Width of Ringwall (ft)	1.6667
Height of Ringwall (ft)	8.7500
Projection of Ringwall Above Grade (ft)	0.5000
Radius to Outside of Ringwall (ft)	13.3333
Radius to Inside of Ringwall (ft)	11.6667
Total Area of Vertical Steel (sqin)	31.4159
Area of Vertical Steel on Each Face (sqin)	15.7080
Total Area of Hoop Steel (sqin)	4.2001
Area of Hoop Steel on Each Face (sqin)	2.1000

FOOTING DESIGN

Outside Diameter of Footing (ft)	34.1875
Inside Diameter of Footing (ft)	12.5000
Width of Footing (ft)	10.8438
Depth to Bottom of Footing (ft)	10.0000
Thickness of Footing at Ringwall (ft)	1.7500
Thickness of Footing at Edge (ft)	1.0000
Total Area of Top Radial Steel (sqin)	19.8198
Total Area of Bottom Radial Steel (sqin)	62.8072
Total Area of Hoop Steel (sqin)	4.9187
Area of Hoop Steel on Each Face (sqin)	2.4594
Bearing Stress Due to D.L. Water and Snow	3.1123
Bearing Stress Due to D.L. Water and Wind	4.9046
Bearing Stress Due to D.L. Water and Seismic	3.9650
Safety Factor Due to Overturning form Wind	2.7558
Safety Factor Due to Overturning from Seismic	5.6316

FOUNDATION QUANTITIES

Quantity of Concrete in Ringwall (cuyd)	42.4221
Quantity of Concrete in Footing (cuyd)	44.1133
Total Quantity of Concrete in Found.(cuyd)	86.5354
Quantity of Ringwall Steel (lb)	2870.0815
Quantity of Footing Steel (lb)	4276.3009
Total Quantity of Foundation Steel (lb)	7146.3824